Goal-oriented DFR

A primal and an adjoint network, trained together

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Goal-Oriented DFR series | Part 2

Where we are

- Part 1: DFR gives a loss, the H^{-1} dual norm of the residual, that is equivalent to the global energy error.
- But we often want a single **quantity of interest** $J(u^*)$, and global control is indirect and wasteful.
- This deck: weight the DFR loss toward J, using the dual-weighted residual (DWR) idea.

A quantity of interest, concretely

J takes the whole solution and returns *one number*, and it must be **linear** in u. The usual choices are:

$$\underbrace{u(x_0)}_{\text{point value}}\,, \qquad \underbrace{\frac{1}{|\omega|}\int_\omega u\,dx\,,}_{\text{average over }\omega} \qquad \underbrace{\int_\Gamma \nabla u\cdot n\,ds}_{\text{flux through }\Gamma}\,.$$

- This work uses the point value, at $x_0 = (0.65, 0.65)$ in 2D.
- In $d \geq 2$, $u \mapsto u(x_0)$ is not bounded on H^1, so we smooth it: $J(u) = \int_\Omega \eta_\sigma(x - x_0)\,u(x)\,dx$, with η_σ a narrow normalized bump ($\sigma = 0.07$ in 2D).
- **The problem:** the DFR loss of Part 1 is global. Nothing in it knows where x_0 is.

Takeaway: One number out of a whole field, and a loss that cannot see it. The adjoint is what tells the loss where to look.

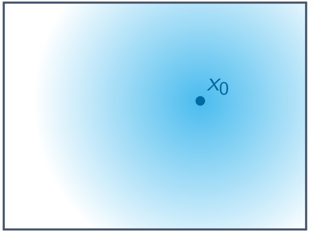
Section 1

The dual-weighted residual idea

The adjoint as an importance map

Classical finite elements answer "where does the residual matter for my QoI?" with an **adjoint** solution z^*.

- z^* solves the **adjoint problem**: same domain, same boundary conditions, same operator, but forced by the QoI rather than by f: for the point value above, $-\Delta z^* = \eta_\sigma(\cdot - x_0)$.
- So z^* is large where a residual most affects $J(u^*)$, and small where it does not.



z^* over the domain Ω

Takeaway: Same operator as the primal problem, different forcing: the QoI takes the place of f, so z^* maps where the residual matters.

The error identity

So far the residual has been a *function*, $f + \Delta u_\theta$. Write it instead as something that eats a test function v, namely $R(u_\theta)(v) = \int_\Omega (f + \Delta u_\theta)\, v\, dx$. Feeding it the adjoint gives the QoI error exactly:

$$J(u^*) - J(u_\theta) \;=\; R(u_\theta)(z^*).$$

- The residual is *weighted by the importance map* before being scored, so it is judged where the QoI cares.
- But z^* is the **true** adjoint: exact, not computable. Replacing it by a network is what Part 3 pays for.

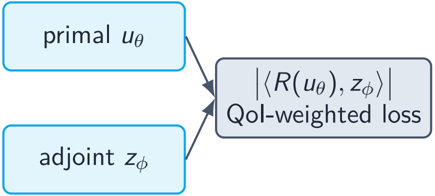
Takeaway: The QoI error is the residual tested against the true adjoint. Part 3 proves it, and prices the substitution.

Section 2

The construction

Two networks

Train a **primal** network and an **adjoint** network together.



- The primal minimizes the QoI-weighted residual, putting its accuracy where the adjoint says it matters.
- The adjoint solves its *own* DFR problem, so it genuinely approximates z^*.

The subtlety that makes it work: a stop-gradient

The goal term $|\langle R(u_\theta), z_\phi \rangle|$ is a single scalar. If the **adjoint** were allowed to descend it too:

- × the optimizer would drive the pairing to zero trivially, by making z_ϕ *orthogonal* to the residual,
- × without z_ϕ approximating the true adjoint at all. The localization is lost.

Fix: hold the adjoint fixed in the goal term (a stop-gradient). The adjoint is trained *only* by its own residual $\|R^*(z_\phi)\|^2_{U^*}$; the primal is guided by the QoI-weighted term.

Takeaway: Stop-gradient stops the adjoint from cheating. The difference between a working method and a degenerate one.

The training objective

$$\mathcal{L}(\theta, \phi) = \underbrace{\left|R(u_\theta)(\bar{z}_\phi)\right|}_{\text{goal term (primal only)}} + \beta \underbrace{\|R(u_\theta)\|_{V^*}^2}_{\text{primal DFR}} + \alpha \underbrace{\|R^*(z_\phi)\|_{U^*}^2}_{\text{adjoint DFR}}$$

- $\bar{z}_\phi$: adjoint held fixed (the stop-gradient).
- The pairing $R(u_\theta)(z_\phi)$ is evaluated on the same grid DFR already uses; no extra quadrature.
- Same spectral evaluation and boundary-cutoff construction as plain DFR. The only new ingredients are the adjoint and the goal term.

The algorithm, in one loop

1. Precompute the DST matrix and the H^{-1} weights.
2. Each step, by DST, form the residual coefficients $\hat{R}(u_\theta)$ and $\hat{R}^*(z_\phi)$.
3. Primal loss $\mathcal{L}_u = |R(u_\theta)(\bar{z}_\phi)| + \beta \, \|R(u_\theta)\|^2_{V^*}$ updates θ.
4. Adjoint loss $\mathcal{L}_z = \alpha \, \|R^*(z_\phi)\|^2_{U^*}$ updates ϕ.
5. Adam, then an L-BFGS polish.

Boundary conditions are enforced by a cutoff: $u_\theta = \varphi \, \tilde{u}_\theta$, with φ vanishing on $\partial\Omega$; likewise for z_ϕ.

Takeaway: Next: what this construction lets us prove.